HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · CONSUMER DURABLES MANUFACTURING · SUPPLY CHAIN

ROOM-AC PLANT · ONE INWARD GATE · SIX DOCKS · ONE FG YARD

Logistics Digital Twin

A 3D twin of the plant follows every supplier truck from gate to dock and every packed lot to dispatch, and lets the plant test peak-season what-ifs before a single truck queues.

~~3:05~~ → **1:55** hours, inbound truck turnaround gate to gate



−38 %
Inbound truck turnaround, gate to gate
▼ 3 h 05 min → 1 h 55 min



91 %
Dock use while trucks are waiting
▲ from 64% last peak season



−52 %
Truck detention charges
▼ peak season vs peak season

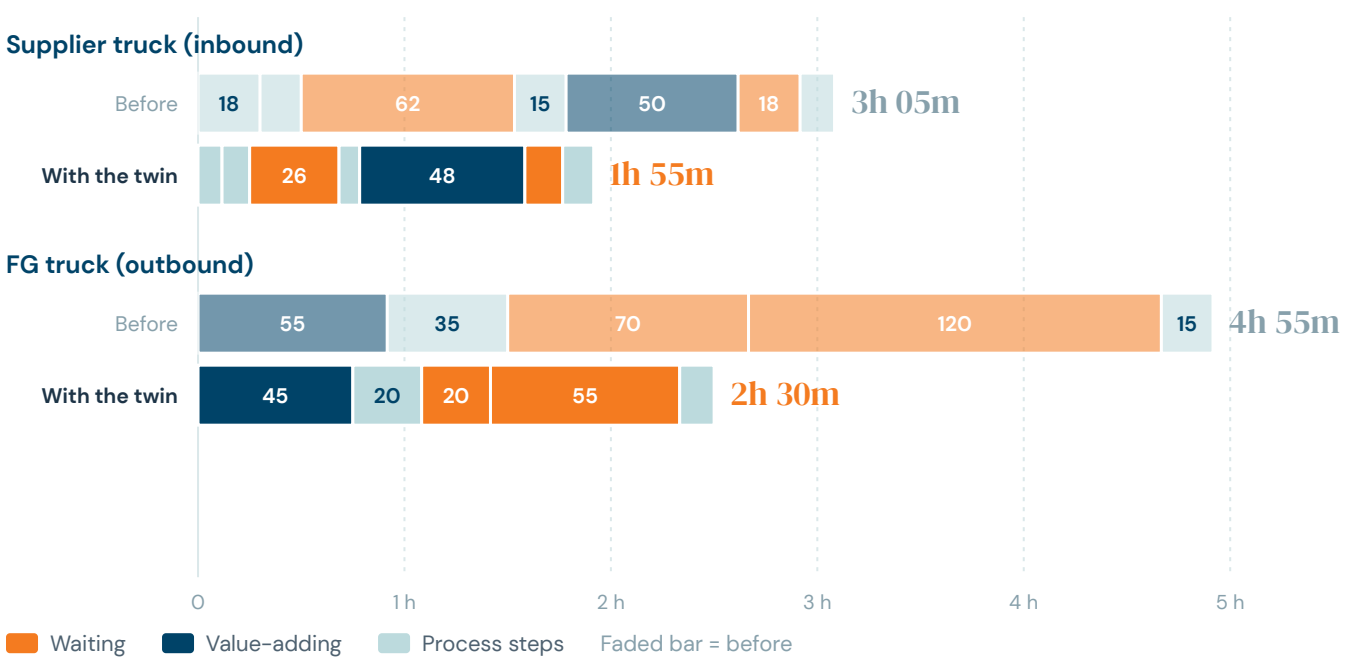


< 2 min
Time to locate any truck on site
▼ from ~20 min of phone calls

Rings: turnaround now as a share of before (inner ring = before); dock use with last season's 64% inside; detention left after the 52% cut; 2 of the old ~20 minutes. Figures indicative; validated metrics available on request.

01 Less waiting, not faster unloading

Anatomy of a truck visit · average minutes per step, peak season



WHERE THE TIME WENT
Waiting, not work
Inbound: parking and GRN holds were 80 of 185 minutes. Outbound: documents and transporters were 190 of 295.

WHAT BARELY MOVED
Unloading: 50 → 48 min
The docks were never slow. The twin removed the waiting around them.

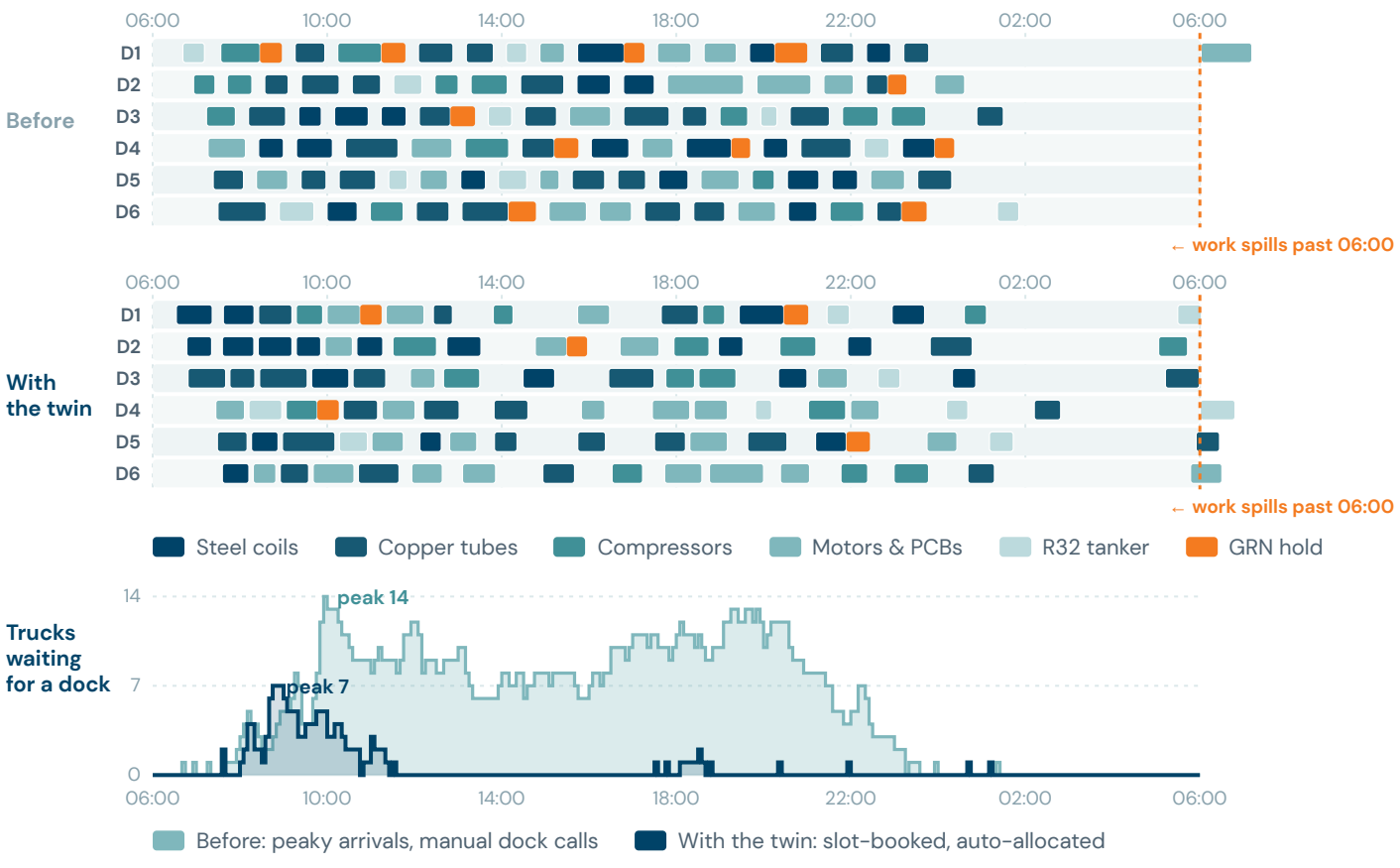
BIGGEST SINGLE CUT
Transporter wait 120 → 55 min
A yard board by truck size and destination, with documents raised at QA.

BUILT FROM THE PLANT'S OWN RECORDS
~600 truck visits time-stamped
Gate registers, weighbridge slips and GRNs; site mapped as 12 parking bays, 6 docks, 24 dispatch bays.

02 One peak day, run twice through the simulator

Dock what-if · every block is one truck at a dock

96 inbound trucks / day · **6** receiving docks · **42 min** average unloading · **12%** trucks held for GRN



Same 96 trucks, same 6 docks
Simulated day, before → with the twin

Average wait for a dock
~~1 h 19 min~~ → **8 min** −90%

90% of trucks wait less than
~~2 h 13 min~~ → **34 min** −74%

Average turnaround, gate to gate
~~2 h 34 min~~ → **1 h 09 min** −55%

Dock use while trucks wait
~~72%~~ → **81%** +9 pts

Peak trucks waiting
~~14~~ → **7** −50%

Detention cost per month
~~₹84k~~ → **₹0** −100%

Indicative discrete-event simulation. "Before" assumes peaky arrivals, 15 min at the gate and manual dock calls; "with the twin" assumes slot-booked arrivals, pre-registered trucks and automatic allocation. Detention estimated at ₹250 per truck-hour beyond 3 hours, 26 working days a month.

03 Four pain points, four fixes

Before → after, as the plant now runs

BEFORE
Queues next to empty docks
Trucks waited outside for hours while two of six docks sat idle. Security saw the queue, stores saw the docks.

AUTO DOCK ALLOCATION

AFTER
Next free dock, automatically
Yard truck called by PA and WhatsApp; the 08:00 dock plan reviewed on the twin.

BEFORE
Four registers, no single truth
Gate pass, weighbridge slip, GRN and dispatch register in different books. One truck took a round of calls.

ONE LIVE SCREEN

AFTER
Every truck on the twin
Cargo, dock and time on site, on screens at security, stores and dispatch.

BEFORE
An overflowing FG yard
Loaded trucks waited for documents, seal checks and transporters. Detention climbed every peak season.

YARD BOARD

AFTER
Documents raised at QA
Yard board by truck size and destination; GRN holds turn the dock red and alert stores in a minute.

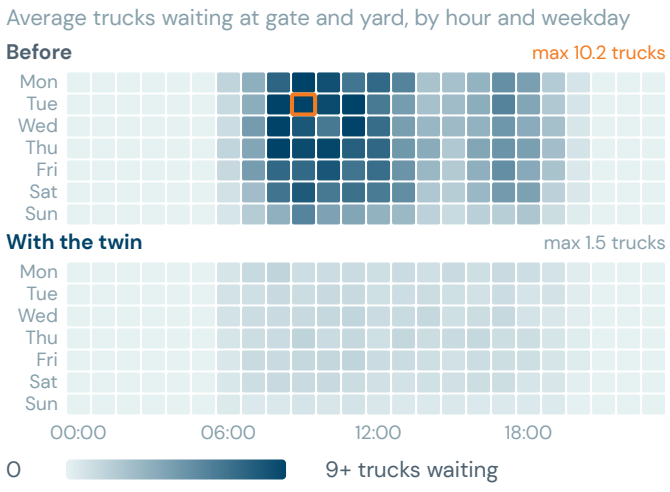
BEFORE
Lost at shift change
Three shifts, verbal handovers. A dock blocked at 13:50 was still blocked at 15:00.

AUTO SHIFT BRIEF

AFTER
The handover writes itself
Queue above 3, dock blocked, yard above 70% or turnaround above 2 h each trigger an action for the next shift.

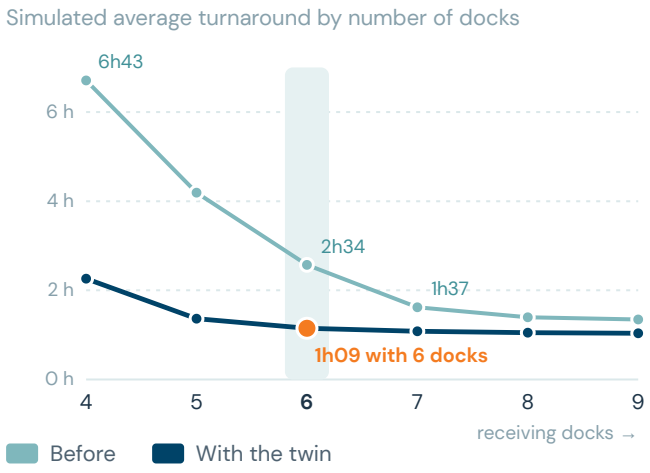
04 When queues build

Average trucks waiting at gate and yard, by hour and weekday



05 How many docks?

Simulated average turnaround by number of docks



With the twin, six docks already turn trucks in about an hour; a seventh adds little. Before, even nine docks still averaged 1 h 21 min.

06 How HMC ran it

2 weeks	**Discover · walk the truck** Followed trucks highway to dock; split ~600 visits into steps to find the waiting.
2 weeks	**Design · model the plant** Zones, dock rules, GRN and shift holds; KPI tree agreed with the plant head.
5 weeks	**Build · a twin you can watch** Three.js plant twin with an event engine; what-if simulator calibrated on visit data.
1 quarter	**Adopt · run the day from it** Screens at security, stores and dispatch; blocked-dock alerts; shift briefs.
∞ ongoing	**Scale · live feeds, more plants** Gate-pass, RFID and weighbridge feeds; supplier slot booking; template for other plants.

What transfers to any plant gate

01 — Make the waiting visible
Most of a truck's visit was invisible waiting, not slow unloading. One screen cut it before any new dock was built.

02 — Simulate the season, then spend
Running peak arrivals through the simulator showed when a seventh dock pays back and when slot booking alone is enough.

03 — Let the twin write the handover
An auto-drafted shift brief kept context across three shifts and gave the plant head one page every morning.

Built with **Three.js** · **Claude** · **Python** (SimPy) · **D3.js** · gate, weighbridge & ERP GRN logs | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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